Sunanda Tata

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Summary

Full-stack AI and software engineer with **3+ years of experience** building enterprise applications, **RAG systems**, and scalable **REST APIs** using **Java, Python, React**, and cloud-native technologies. **M.S. Computer Science** candidate with proven impact improving workflow efficiency, retrieval quality, system reliability, and data-platform performance.

Education

California State University, Fullerton — Fullerton, CA
Master of Science in Computer Science — Aug 2024 – May 2026

Technical Skills

Programming & Full-Stack: Python, Java, TypeScript, JavaScript, Node.js, React.js, FastAPI, Express.js, Spring Boot, Spring WebFlux, RESTful APIs, SQL
Generative AI & Agents: Generative AI, Large Language Models (LLMs), Machine Learning, Natural Language Processing (NLP), LangChain, LLM APIs, AI Agents, Prompt Engineering.
RAG & Enterprise Search: Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search, Vector Search, Reranking, Vector Databases, Knowledge Graphs, Qdrant, Neo4j, BM25
Cloud, Data & Reliability: AWS, Azure, Google Cloud Platform, Docker, Kubernetes, Apache Kafka, Redis, CI/CD, Jenkins, Debugging, Code Review, Cross-Functional Collaboration, Groundedness Evaluation, AI Guardrails, Human-in-the-Loop Workflows, Unit Testing, Integration Testing, TDD

Professional Experience

Pacific Life — Newport Beach, CA
Software Developer Intern — Jun 2025 – Mar 2026

- Reduced **LLM inference costs by 28%** by developing a **Python/FastAPI semantic routing service** that directed **70% of routine underwriting requests** to lightweight models, reserving larger models for complex risk evaluations.
- Maintained **99.9% underwriting workflow availability** by hosting fault-tolerant backend services on **Amazon EC2**, using **AWS Lambda** for event-driven tasks, and implementing **CloudWatch monitoring**, automated retries, circuit breakers, and cache fallbacks.
- Simplified property-risk assessments, reducing **manual review effort by 75%**, by integrating **Esri REST APIs** with **AWS Lambda** workflows to automate environmental-hazard evaluations and return structured risk results to underwriting services.
- Accelerated underwriting guideline retrieval from **15 minutes to 30 seconds per application** by building a **Python-based RAG pipeline** with **Amazon S3** document ingestion, vector search, and source-grounded REST API responses.
- Expedited underwriting referral resolution from **8 to 3 minutes per case** by developing **REST APIs and a React/TypeScript workbench** using **Spec-Driven Development**, documented API contracts, and acceptance criteria; authored internal technical wikis and troubleshooting guides.

Hitachi Vantara — Bengaluru, India
Software Engineer — Aug 2022 – Aug 2024

- Expanded metadata indexing to **1M+ enterprise assets** by developing scalable **Java/Spring Boot REST APIs** for Lumada Data Catalog, integrating automated metadata harvesting and schema discovery across databases and data lakes.
- Improved cross-environment metadata synchronization speed by **45%** by publishing metadata-change events through **Apache Kafka**, processing batch synchronization with **Spring Batch**, and using **Amazon S3** for intermediate data storage.
- Protected **99.9% ingestion-pipeline availability** by operating **Java microservices on Amazon EC2**, implementing retry queues and fault-tolerant event processing, and participating in **on-call rotations** to troubleshoot production incidents.
- Achieved **40% lower metadata search latency** by migrating blocking REST endpoints to non-blocking **Spring WebFlux microservices** and developing **React dashboards** for real-time asset-hierarchy visualization.
- Delivered **60% faster regression-testing cycles** by automating **Jenkins CI/CD pipelines** for **Docker-containerized microservices on Amazon EKS**, integrating **CloudWatch monitoring**, and authoring internal deployment wikis and incident-response runbooks.

Open Source Contributions

RAGFlow — github.com/sunandatata/ragflow
Open Source Contributor

- Improved **retrieval API correctness across 20+ regression scenarios** by fixing parameter precedence for top-k retrieval, similarity thresholds, reranking, and saved search configurations.
- Restored **correct top-k ordering across 25+ reranking test cases** by sorting reranked chunks before pagination, preventing high-relevance results from being excluded from LLM context.
- Improved **multilingual retrieval consistency across 3 languages** by extending Agent memory retrieval with cross-language support and validating dataset and memory behavior through **15+ regression tests**.

Projects

ChronoMind – Enterprise RAG & Vector Search Engine

- Engineered a hybrid retrieval engine combining dense vector search, sparse BM25 retrieval, and knowledge-graph traversal using **Qdrant, Neo4j, LangChain, and cross-encoder reranking**.
- Outperformed vector-only baselines by **18% in Recall@10** and **0.14 in MRR** across multi-hop reasoning benchmarks by fusing semantic vector similarity, keyword matching, and Neo4j entity-relation paths.
- Optimized retrieval latency under **120ms** across 100k+ documents by caching frequent query subgraphs in Redis and running parallel sparse/dense candidate generation prior to reranking.

Ride Matching Platform github.com/sunandatata/Ride-Matching-Platform

- Built a resilient real-time ride-dispatch platform using **Node.js, Express, TypeScript, and React**, separating 7 backend domains to isolate location tracking from dispatch workflows.
- Prevented race conditions where multiple drivers accept the same ride by implementing **atomic state transitions with distributed locks**, rejecting stale concurrent requests with sub-millisecond overhead.
- Ensured fault-tolerant client synchronization by engineering **WebSocket heartbeat reconnects with state hydration** for network drops and **idempotency keys** to safeguard retried booking and payment requests against duplicate execution.
- Verified end-to-end distributed state consistency under network partitions and high concurrency through **11+ automated Docker-based integration test suites**.

AI2-SED Defense Framework

- Built an adversarial phishing-defense pipeline coupling a **fine-tuned RoBERTa classifier** and URL heuristic model with a **T5 generative attack engine** to simulate zero-day evasion tactics.
- Raised adversarial attack robustness by **22% against semantic and lexical perturbations** through iterative red-team adversarial training and confidence-calibrated ensembling.
- Deployed a low-latency **FastAPI inference service** and Streamlit analytics dashboard with structured JSON telemetry to inspect prediction drift and model explainability.